$+ \quad \times$

$- \quad \div$

$$\{|n\rangle\} \longrightarrow G\left(\sum_{n=1}^{d} a_n|n\rangle\right) = \sum_{n=1}^{d} a_n G(|n\rangle)$$

YOU CAN REPRESENT AN OPERATOR AS A MATRIX

$$\begin{array}{c|c} & |1\rangle \quad \cdots \quad |d\rangle \\ \langle 1| \\ \vdots \\ \langle d| & \begin{pmatrix} G_{11} & \cdots & G_{1d} \\ \vdots & \ddots & \vdots \\ G_{d1} & \cdots & G_{dd} \end{pmatrix} \end{array} \qquad G_{mn} = \langle m|G|n\rangle$$

FOR THE PAULI OPERATOR Y AND THE $\{|0\rangle, |1\rangle\}$ BASIS WE HAVE

$$Y = -i|0\rangle\langle 1| + i|1\rangle\langle 0| \qquad \begin{array}{c|cc} & |0\rangle & |1\rangle \\ \langle 0| & 0 & -i \\ \langle 1| & i & 0 \end{array}$$

$$\langle 1|Y|0\rangle = -i\underbrace{\langle 1|0\rangle}_{0}\underbrace{\langle 1|0\rangle}_{0} + i\underbrace{\langle 1|1\rangle}_{1}\underbrace{\langle 0|0\rangle}_{1} = i$$

GIVES AS RESULT A COMPLEX NUMBER

LINEAR FUNCTIONAL

THE TRACE IS A LINEAR FUNCTIONAL ON OPERATORS $\mathrm{TR}: O \to \mathbb{C}$

$$\mathrm{Tr}\,A = \sum_n \langle n|A|n\rangle = \sum_n A_{nn}$$

IN QUANTUM MECHANICS, AN OBSERVABLE IS A PROPERTY OF THE SYSTEM WHOSE

SUPPOSE A_n IS THE NUMERICAL VALUE ASSOCIATED WITH THE nTH OUTCOME, WHICH HAS A BASIS ELEMENT $|n\rangle$. THE OPERATOR A ASSOCIATED WITH THE OBSERVABLE A IS DEFINED BY

$$A|n\rangle = A_n|n\rangle, \ \forall n \qquad A = \sum_n A_n |n\rangle\langle n|$$

IF THE SYSTEM IS IN A GENERIC STATE $|\psi\rangle$ WE TALK ABOUT THE EXPECTATION VALUE

$$\langle A\rangle = \langle\psi|A|\psi\rangle = \langle\psi|\left(\sum_n A_n|n\rangle\langle n|\right)|\psi\rangle = \underbrace{\sum_n A_n P(n)}_{\text{WEIGHTED AVERAGE}}$$

EXPECTATION VALUE

WE CAN ALSO WRITE THE EXPECTATION VALUE OF A IN TERMS OF THE REPRESENTATION OF THE STATE AND THE OPERATOR

$$\langle A\rangle = \psi^\dagger A \psi$$

EXAMPLE: CALCULATE $\langle S_x\rangle$ OF THE STATE $|\psi\rangle = 0.6|z_+\rangle - 0.8|z_-\rangle$

NOT IN THE EXAM

! BE SURE THAT THE STATE IS NORMALIZED

$|\langle\psi|\psi\rangle|^2 = P_\psi(\psi) = 1$

1. CHECK THAT IS NORMALIZED

$B(|z_+\rangle, |z_-\rangle)$ $\quad \langle\psi|\psi\rangle = 0.6\langle z_+| - 0.8\langle z_-|\,(0.6|z_+\rangle - 0.8|z_-\rangle) =$

$$= 0.36\underbrace{\langle z_+|z_+\rangle}_{1} - 0.48\underbrace{\langle z_-|z_+\rangle}_{0} - 0.48\underbrace{\langle z_+|z_-\rangle}_{0} + 0.64\underbrace{\langle z_-|z_-\rangle}_{1} = 0.36 + 0.64 = 1$$

NORMALIZED :)

OPTION 1

2. UNDERSTAND WHAT IS S_x – USE REPRESENTATION

$$S_x = \frac{\hbar}{2} X \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\Rightarrow S_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

RULE: $\langle S_x\rangle = \psi^T S_x \psi$

$$\Rightarrow \langle S_x\rangle = (0.6 \ \ -0.8)\ \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0.6 \\ -0.8 \end{pmatrix}$$

$$= \frac{\hbar}{2}(0.6 \ \ -0.8)\begin{pmatrix} -0.8 \\ 0.6 \end{pmatrix} = \frac{\hbar}{2}\left(-0.48 + (-0.48)\right) = \boxed{0.96}\,(-\hbar/2)$$

$0 \le P \le 1$

96% $-\hbar/2$

4% $+\hbar/2$

(2.B) OPTION B $\langle S_x \rangle$ AS OPERATOR

$$S_x = \frac{\hbar}{2}|z_+\rangle\langle z_-| + \frac{\hbar}{2}|z_-\rangle\langle z_+| = \langle\psi|\frac{\hbar}{2}|z_+\rangle\langle z_-| + \frac{\hbar}{2}|z_-\rangle\langle z_+|\psi\rangle$$

$[\hbar = J\cdot s]$

$$|\langle\psi|z_+\rangle|^2 \qquad \langle S_x\rangle = \frac{\hbar}{2}P(z_+) + \left(-\frac{\hbar}{2}\right)P(z_-)$$

GIOVANN

EXERCISE 2:

Exercise

Exercise 3.32 A two-level atom has possible energies E_0 and E_1. Write down the Hamiltonian H as a sum of outer products. Now suppose that the atom is in the state $|\psi\rangle = \left(\frac{1}{2}\right)|E_0\rangle + \frac{\sqrt{3}}{2}|E_1\rangle$. Use your operator H to find the expectation value of the atom's energy

E_1 ———— $|E_1\rangle$

E_0 ———— $|E_0\rangle$

① WRITE THE ENERGY OPERATOR H

$$H|E_0\rangle = E_0|E_0\rangle$$
$$H|E_1\rangle = E_1|E_1\rangle$$

$\boxed{\text{IF } A|N\rangle = A_N|N\rangle \Rightarrow A = \sum_N A_N|N\rangle\langle N|}$

$\Downarrow$

WRITE H AS:

$$H = E_0|E_0\rangle\langle E_0| + E_1|E_1\rangle\langle E_1|$$

TEST 1 ✓ :)

$$H|E_0\rangle = (E_0|E_0\rangle\langle E_0| + E_1|E_1\rangle\langle E_1|)|E_0\rangle = E_0|E_0\rangle$$

TEST 2 ✓ :)

$$H|E_1\rangle = E_1|E_1\rangle = (\text{" " "})\text{" "} = E_1|E_1\rangle$$

SYMMETRIC $\Rightarrow$ WE DEFINED THE OPERATOR

② FIND THE EXPECTATION VALUE OF THE ENERGY FOR THE STATE $|\psi\rangle = \frac{1}{2}|E_0\rangle + \frac{\sqrt{3}}{2}|E_1\rangle$

(2.1) CHECK THE STATE IS NORMALIZED

$$\langle\psi|\psi\rangle = \left(\frac{1}{2}\langle E_0| + \frac{\sqrt{3}}{2}\langle E_1|\right)\left(\frac{1}{2}|E_0\rangle + \frac{\sqrt{3}}{2}|E_1\rangle\right) = \frac{1}{4} + \frac{3}{4} = 1 \quad :)$$

(2.2)

$$\langle H\rangle = \langle\psi|H|\psi\rangle \qquad H = E_0|E_0\rangle\langle E_0| + E_1\rangle\langle E_1|)$$

$$H = \begin{pmatrix} E_0 & 0 \\ 0 & E_1 \end{pmatrix} \qquad \langle E_0|H|E_1\rangle = \langle E_0|E_1|E_1\rangle = E_1\langle E_0|E_1\rangle \underset{0}{=}$$

$$\langle H\rangle = \langle\psi|H|\psi\rangle = \begin{pmatrix}\frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix}\begin{pmatrix} E_0 & 0 \\ 0 & E_1 \end{pmatrix}\begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix} = \begin{pmatrix}\frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix}\begin{pmatrix} \frac{1}{2}E_0 \\ \frac{\sqrt{3}}{2}E_1 \end{pmatrix} = \frac{1}{4}E_0 + \frac{3}{4}E_1$$

EXPECTATION VALUE

$$\langle A^K\rangle = \langle\psi|A^K|\psi\rangle \qquad A^K = \sum_N (A_N)^K|N\rangle\langle N|$$

EX: CONSIDER A SPIN-1/2 PARTICLE IN THE STATE $|x_+\rangle$. CALCULATE $\langle S_z\rangle^2$ AND $\langle S_z^2\rangle$

$$|\psi\rangle = |x_+\rangle \qquad S_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\frac{\hbar}{2}$$

$$\{|z_+\rangle, |z_-\rangle\}$$

$$\hat{S}_z = \frac{\hbar}{2}|z_+\rangle\langle z_+| - \frac{\hbar}{2}|z_-\rangle\langle z_-| \qquad |x_+\rangle = \frac{1}{\sqrt{2}}(|z_+\rangle + |z_-\rangle) \qquad |x_+\rangle = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

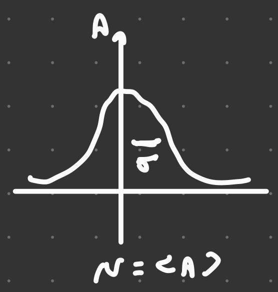
$$\langle S_z\rangle_{x_+}$$

(THEN SQUARE IT) $\langle S_z \rangle_{x_+} = \langle \psi | S_z | \psi \rangle = \langle x_+ | S_z | x_+ \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0 \Rightarrow \langle S_z \rangle^2 = 0$

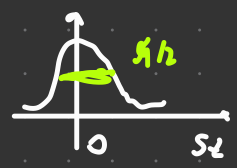
CALCULATE $\langle S_z^2 \rangle$

$$S_z^2 = \left(\frac{\hbar}{2} z\right)^2 = \frac{\hbar^2}{4} z^2 = \frac{\hbar^2}{4} I \Rightarrow z^2 = 1 \qquad S_z^2 = \frac{\hbar^4}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\langle \psi | S_z^2 | \psi \rangle = \langle \psi | \frac{\hbar^2}{4} I | \psi \rangle = \frac{\hbar^2}{4} \langle \psi | \overbrace{I | \psi \rangle}^{\text{IS } \psi \text{ (IS THE IDENTITY}} = \frac{\hbar^2}{4}$$



$$\sigma^2 = \langle A^2 \rangle - \langle A \rangle^2$$



DENSITY OPERATOR

WE INTRODUCE THE DENSITY OPERATOR OF THE STATE $|\psi\rangle$ AS:

$$\rho = |\psi\rangle\langle\psi|$$

WE HAVE THAT

$$\langle A \rangle = TR(\rho A)$$

WE KNOW

$$\boxed{\begin{aligned} \rho &= |\psi\rangle\langle\psi| \\ A &= \sum_n a_n |n\rangle\langle n| \end{aligned}}$$

$$TR(\rho A) = \sum_n \langle n | \rho A | n \rangle = \sum_n \langle n | \psi \rangle \langle \psi | A | n \rangle$$

$$= \sum_n \langle n | \psi \rangle \langle \psi | a_n | n \rangle = \sum_n a_n \underbrace{\langle n | \psi \rangle \langle \psi | n \rangle}_{P(n)}$$

$$= \sum_n a_n P(n) = \langle A \rangle$$

ADJOINT OPERATOR

IF A IS AN OPERATOR IN A HILBERT SPACE $\mathcal{H}$ ONE CAN DEFINE THE ADJOINT OPERATOR $A^\dagger$

$$\langle \alpha | A^\dagger | \beta \rangle = (\langle \beta | A | \alpha \rangle)^*$$

$$\Downarrow$$

$$(A^\dagger)_{mn} = A^*_{nm}$$

NOTE THAT • $(AB)^\dagger = B^\dagger A^\dagger$

• $|\beta\rangle = A|\alpha\rangle \Leftrightarrow \langle \beta | = \langle \alpha | A^\dagger$

WE CALLED HERMITIAN OPERATOR, THE OPERATOR THAT SATISFY THE FOLLOWING RELATION

$$A^\dagger = A$$

H IS HERMITIAN

$$H = \begin{pmatrix} E_0 & 0 \\ 0 & E_1 \end{pmatrix} \qquad H^\dagger \begin{pmatrix} E_0 & 0 \\ 0 & E_1 \end{pmatrix}^t = \begin{pmatrix} E_0 & 0 \\ 0 & E_1 \end{pmatrix}$$

$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow S_z^\dagger = S_z$$

$$S_y = \frac{\hbar}{2}\begin{pmatrix}0 & i\\ -i & 0\end{pmatrix} \Rightarrow S_y^\dagger = \frac{\hbar}{2}\left[\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\right]^* = \frac{\hbar}{2}\begin{pmatrix}0 & i\\ -i & 0\end{pmatrix} = S_y$$ S_y IS HERMITIAN

FROM NOW ON, WE SHALL ASSUME THAT ANY OBSERVABLE IS ASSOCIATED WITH AN HERMITIAN OPERATOR

ONE CAN PROVE THAT IF A IS HERMITIAN, THEN $\langle A\rangle \in \mathbb{R}$

$A = A^\dagger \Leftrightarrow A$ IS HERMITIAN

TO PROVE → $\langle A\rangle$ IS REAL

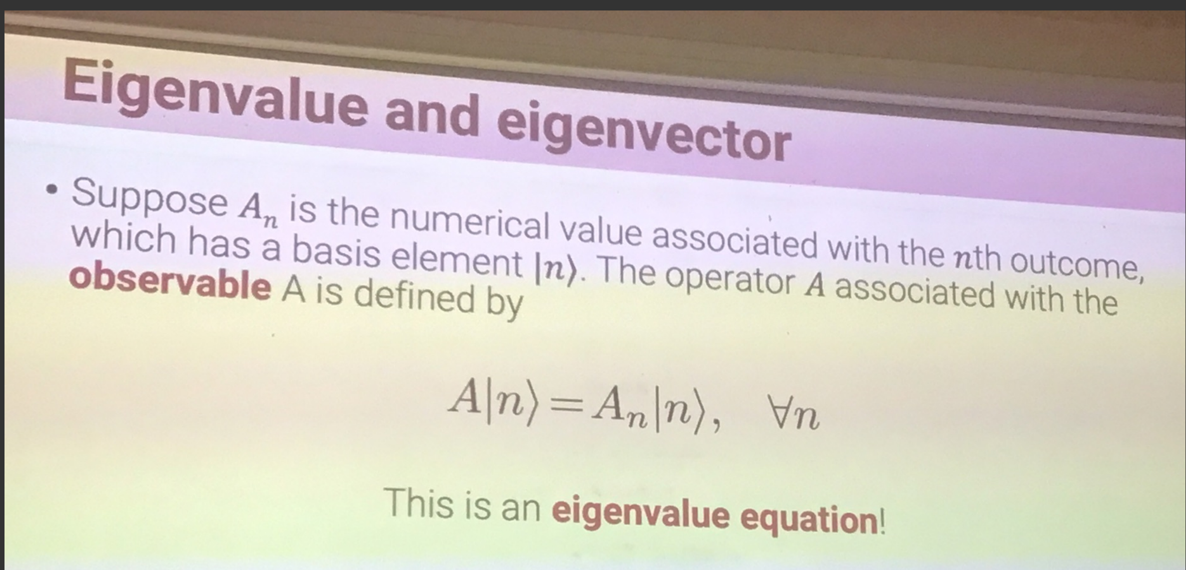
$$\langle A\rangle = \langle\psi|A|\psi\rangle = \langle\psi|A^\dagger|\psi\rangle = (\langle\psi|A|\psi\rangle)^* = (\langle A\rangle)^*$$ IF $z = z^* \Rightarrow z \in \mathbb{R}$

AN OPERATOR U IS UNITARY IF ITS ADJOINT EQUALS ITS INVERSE

$$U^\dagger U = UU^\dagger = 1$$

WE SAW THAT THE ACTION OF UNITARY OPERATORS PRESERVES THE INNER PRODUCT OF TWO VECTORS

EIGENVALUE AND EIGENVECTOR



THIS MEANS THAT ALL THE PROPERTIES OF EIGENVALUES PROBLEMS APPLY, INCLUDING SPECTRAL THEOREM

FOR ANY NORMAL OPERATOR A, WE CAN FIND A UNITARY MATRIX U SUCH THAT

$$A = UDU^\dagger$$

WHERE D IS DIAGONAL MATRIX

NORMAL N $\quad NN^\dagger = N^\dagger N$

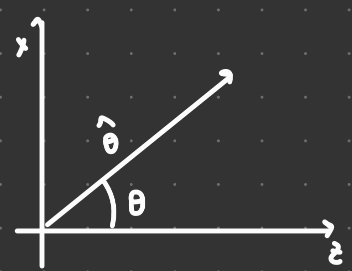
$$NN^\dagger - N^\dagger N = 0 = [N, N^\dagger] = 0$$

$[A, A^\dagger] = [A, A] = 0$ HERMITIAN OPERATOR ARE NORMAL $\quad$ HERMITIAN $\Rightarrow$ NORMAL

COROLLARY IF A IS HERMITIAN, THEN D IS A REAL MATRIX

WE KNOW THAT THE SPIN OPERATORS ASSOCIATED WITH THE PAULI MATRICES ARE

$$S_x = \frac{\hbar}{2}X \qquad S_y = \frac{\hbar}{2}Y \qquad S_z = \frac{\hbar}{2}Z$$



$$\hat\theta = \sin\theta\,\hat x + \cos\theta\,\hat z$$
$$S_\theta = \sin\theta\, S_x + \cos\theta\, S_z$$

$$S_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

1) WHAT ARE THE POSSIBLE OUTCOMES OF A SPIN MEASUREMENT ALONG θ?

2) WHAT ARE THE STATES ASSOCIATED WITH THESE OUTCOMES?

1) $S_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} \hbar/2$

$A|\psi_\alpha\rangle = \alpha|\psi_\alpha\rangle \Rightarrow \underbrace{(A-\alpha I)}|\psi_\alpha\rangle = 0$

$\hookrightarrow \neq 0$

$(A')\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = 0 \rightarrow$ TRIVIAL SOLUTION: $\underline{x} = \underline{0}$ (BUT $\psi_\alpha \neq 0$)

$\det(S_\theta - \lambda I) = 0$

$$\det\begin{pmatrix} \hbar/2\cos\theta - \lambda & \hbar/2\sin\theta \\ \hbar/2\sin\theta & -\hbar/2\cos\theta - \lambda \end{pmatrix} = (\hbar/2\cos\theta - \lambda)(-\hbar/2\cos\theta - \lambda) - \hbar/2\sin^2\theta = -(\hbar/2\cos\theta - \lambda)(\hbar/2\cos\theta + \lambda) -$$

$$- \hbar^2/4\sin^2\theta = -\frac{\hbar^2}{4}\cos^2\theta + \lambda^2 - \frac{\hbar^2}{4}\sin^2\theta$$

$$= \lambda^2 - \hbar^2/4\underbrace{(\cos^2\theta + \sin^2\theta)}_{\hookrightarrow 1} \overset{\text{DEF}}{=} 0 \Rightarrow \lambda^2 = \hbar^2/4 \Rightarrow \lambda_\pm = \pm\hbar/2$$

$$\begin{pmatrix} \frac{\hbar}{2}\cos\theta - \hbar/2 & \hbar/2\sin\theta \\ \frac{\hbar}{2}\sin\theta & -\frac{\hbar}{2}\cos\theta - \hbar/2 \end{pmatrix}\begin{pmatrix} \alpha_+ \\ \alpha_- \end{pmatrix} = 0$$

$\lambda_+ = \hbar/2$ $\quad |\lambda_+\rangle = \alpha_+|z_+\rangle + \alpha_-|z_-\rangle$

$\alpha_+(\cancel{\hbar/2}\cos\theta - \cancel{\hbar/2}) + \alpha_-\cancel{\hbar/2}\sin\theta = 0$

$\alpha_+ = -\sin\theta$

$\alpha_- = \cos\theta - 1$

$\alpha_+(\cos\theta - 1) + \alpha_-\sin\theta = 0$

$$\begin{pmatrix} -\sin\theta \\ \cos\theta - 1 \end{pmatrix} \quad (-\sin\theta \quad \cos\theta - 1)\begin{pmatrix} -\sin\theta \\ \cos\theta - 1 \end{pmatrix} = \sin^2\theta + (\cos\theta - 1)^2$$

$N = \sin\theta^2 + (\cos\theta - 1)^2$

$$\begin{pmatrix} \alpha_+ \\ \alpha_- \end{pmatrix} = \frac{1}{N}\begin{pmatrix} -\sin\theta \\ \cos\theta - 1 \end{pmatrix} \qquad |\lambda_+\rangle = \frac{1}{N}\left[-\sin\theta\,|z_+\rangle + (\cos\theta - 1)|z_-\rangle\right]$$

$|\lambda_-\rangle$